

MODULE 03 REFLECTIVE JOURNAL

ML WORKFLOW AND LEARNING TYPES

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LEARNING PROCESS AND CONCEPTUAL UNDERSTANDING

This lab changed how I understand machine learning. Before completing it, I mainly associated machine learning with selecting an algorithm and running code. By working through the Wine classification project in order, I learned that machine learning is a connected workflow: define the problem, explore and prepare the data, select features, split the data, train models, evaluate their performance, and interpret the results. Each stage affects the next one, so a strong model depends on more than the algorithm alone.

The distinction among supervised, unsupervised, and reinforcement learning became clearer when I connected each type to the information available during learning. Supervised learning has labeled examples, so the model has an answer key during training. The Wine project is supervised because chemical measurements are used to predict known wine classes. Unsupervised learning has no labeled answers and searches for patterns or groups. Reinforcement learning improves decisions through interaction, rewards, and penalties. The scenario assessment helped me apply these definitions instead of only memorizing them.

UNDERSTANDING THE DATA AND WORKFLOW

Exploratory Data Analysis showed me why data must be understood before modeling. I examined 178 samples, 13 features, three target classes, and confirmed that there were no missing values. The class-distribution chart showed that the classes were reasonably balanced, while the correlation heatmap showed relationships among the first six features. This connected the Pandas, NumPy, Matplotlib, and Seaborn tools from Module 2 to a practical machine learning workflow.

The data was divided into 142 training samples and 36 testing samples. I learned that the training set teaches the model, while the testing set evaluates performance on unseen data. The use of **stratify=y** preserved similar class proportions in both sets, and **random_state=42** made the experiment reproducible. This helped me understand why evaluating a model on its training data could be misleading and how a separate test set reduces that risk.

MODEL EVALUATION AND CHALLENGES

Logistic Regression achieved 88.9% accuracy, compared with 83.3% for the Decision Tree, so Logistic Regression was the best original model. I had assumed that a flexible model such as a Decision Tree might always perform better, but this comparison showed that performance depends on the data, selected features, parameters, and test split. The Decision Tree was limited to a maximum depth of three, while the relationships in the selected features may have suited Logistic Regression better.

The most challenging part was interpreting the classification report and confusion matrix. Accuracy gives an overall result, but it does not show which classes were confused. The confusion matrix showed that the model classified all class_0 samples correctly but made errors between class_1 and class_2. I therefore learned why precision, recall, and F1-score are useful in addition to accuracy.

EXPERIMENTATION, APPLICATION, AND NEXT STEPS

I experimented with alcohol, hue, and proline as a new three-feature combination. It achieved 100% accuracy on the current test set, demonstrating that feature selection can strongly affect results. However, I also recognized that perfect performance on only 36 test samples does not guarantee perfect performance on future data. I would next apply feature scaling, cross-validation, and parameter tuning, and compare additional algorithms to determine whether the improvement is consistent.

Education is the application area that interests me most. A supervised model could use test scores, homework results, attendance, learning progress, and classroom performance to identify students who may need additional support. Such a model should assist teachers rather than replace professional judgment, and student information must be handled responsibly. My main takeaway is that machine learning is an iterative, evidence-based process. I would like to learn more about overfitting, systematic feature selection, cross-validation, and responsible model deployment.